

071306T4EEN

Electrical Engineering (Power Option) Level 6

ENG/CU/PO/CR/02/6

Install Electrical Power Lines

July/August 2023

Time: 3 Hours



**TVET CURRICULUM DEVELOPMENT, ASSESSMENT AND CERTIFICATION
COUNCIL (TVET CDACC)**

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WRITTEN ASSESSMENT

3 HOURS

INSTRUCTIONS TO CANDIDATE:

*This paper consists of **TWO** sections: **A** and **B**.*

*Answer **ALL** questions in sections A and B in the answer booklet provided.*

Marks for each question are indicated in brackets

Do not write on this question paper.

*Answer the questions in **English**.*

This paper consists of three (3) printed pages.

**Candidate should check the question paper to ascertain that all the pages are
printed as indicated and that no questions are missing.**

SECTION A (40 MARKS)

Answer ALL the questions in this section.

1. Outline **four** considerations for high voltage transmission lines. (4 marks)
2. State **four** effects of vibration on the transmission lines. (4 marks)
3. List any four elements of a high-voltage transmission line. (4 marks)
4. State **four** causes of high voltage surges in transmission lines. (4 marks)
5. Describe the following types of electrical transmission power lines: (4 marks)
 - a) Short transmission lines;
 - b) Long transmission lines.
6. Explain **two** advantages of Miniature Circuit breakers (MCBs) over fuses. (4 marks)
7. Define the following terms as used in protective devices of electrical machines. (4marks)
 - a) Current rating;
 - b) Fusing factor.
8. Overhead conductor depends on tensions, list any **four**. (4 marks)
9. State and explain **two** essential requirements of a switch gear. (4 marks)
10. State **four** ways of reducing corona effect in a transmission lines. (4 marks)

SECTION B (60 MARKS)

Answer any **three** questions in this section.

11. (a) Outline **four** properties of line support in transmission lines. (4 marks)
- (b) Discuss **three** types of insulators used in overhead transmission lines. (6 marks)
- (c) A 3-phase transmission line is being supported by three-disc insulators. The potentials across top unit (i.e., near to the tower) and middle unit are 8 kV and 11 kV respectively. Calculate the:
- i. ratio of capacitance between pin and earth to the self-capacitance of each unit;
 - ii. line voltage;
 - iii. string efficiency. (10 marks)
12. (a) Discuss the following terms as applied to transmission lines:
- i. Voltage regulation
 - ii. Transmission efficiency (4 marks)
- (b) With aid of equations differentiate between Positive and Negative phase sequence (4 marks)
- (c) Using vector diagrams explain the **three** commonly used methods of medium transmissions lines. (12 marks)
13. (a) Explain the **three** constants of a power transmission line. (6 marks)
- (b) Define 'skin effect' and list four factors it depends on. (6 marks)
- (c) The phase voltages across a certain unbalanced three-phase load are given as; $E_R = 176 - j132$, $E_Y = -128 - j96$ and $E_B = -160 + j100$. determine the positive, negative and zero-sequence components for the above voltages. (8 marks)
14. (a) Explain three types of tests carried out in electrical transmission power lines. (6 marks)
- (b) Explain 'corona effect'. (3 marks)
- (c) A three-phase, 220 kV, 50 Hz, 100 km long overhead line consists of three stranded aluminium conductors spaced 5 m apart in equilateral formation. The surrounding air is at a temperature of 27°C and a pressure of 740 mm of mercury. The breakdown strength of air is 3000 V/mm (Max.). Determine:
- (i) The disruptive critical voltage;
 - (ii) The visual critical voltage;
 - (iii) The corona power loss of the line.
- Assume the conductor diameter of 21 mm, the irregularity factor of 0.9 for the disruptive critical voltage and 0.8 for the visual critical voltage. (11 marks)

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